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Probabilistic Action Prediction of Human Drivers Using Dynamic Bayesian Networks

Master Thesis 1111

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List of Abbreviations

AV Autonomous Vehicle

HDV Human-Driven Vehicle

DBN Dynamic Bayesian Network

BN Bayesian Network

DAG Directed Acyclic Graph

MAP Maximum A Posteriori

AI Artificial Intelligence

THW Time Headway

TTC Time to Collision

KI Künstliche Intelligenz

CPU Central Processing Unit

V2X Vehicle-to-Everything

RV Random Variable

CPD Conditional Probability Distribution

CPT Conditional Probability Table

HMM Hidden Markov Model

EM Expectation-Maximization

MLE Maximum Likelihood Estimation

ADAS Advanced Driver Assistance Systems

IDM Intelligent Driver Model

FVDM Full Velocity Difference Model

OVM Optimal Velocity Model

GMM Gaussian Mixture Model

GMR Gaussian Mixture Regression

GPR Gaussian Process Regression

MOBIL Minimizing Overall Braking Induced by Lane Changes

DL Deep Learning

SVM Support Vector Machine

kNN k-Nearest Neighbor

RNN Recurrent Neural Network

LSTM Long Short-Term Memory

NN Neural Network

GNN Graph Neural Network

T-GNN Temporal Graph Neural Network

GCN Graph Convolutional Network

MTR Motion Transformer

HiVT Hierarchical Vector Transformer

PCA Principal Component Analysis

CVAE Conditional Variational Autoencoder

VAE Variational Autoencoder

PECNet Predicted Endpoint Conditioned Network

IRL Inverse Reinforcement Learning

MaxEnt IRL Maximum Entropy Inverse Reinforcement Learning

POMDP Partially Observable Markov Decision Process

PoE Product of Experts

Glossary

Autonomous Vehicle (AV) A vehicle that can drive by itself without human control. It uses sensors and artificial intelligence to understand the surrounding environment and make driving decisions.

Belief State The probability distribution over the latent states given the observed data at a specific time step.

Confusion Matrix A matrix showing the performance of a classification model, across the true and predicted classes.

Directed Acyclic Graph (DAG) A graph made up of nodes and directed edges between them, where the edges do not form any closed loops.

DBN A probabilistic graphical model that uses a directed acyclic graph to represent time-series data, where each node represents a random variable and the edges show how these variables depend on each other within and across time steps.

Emission Model The probability distribution that describes how observations are generated from latent states.

Entropy A measure of uncertainty in a probability distribution. It also quantifies the amount of information needed to describe the distribution.

F1 Score The harmonic mean of precision and recall.

Human-Driven Vehicle (HDV) A vehicle that is controlled by a human driver who performs all driving tasks such as steering, accelerating, and braking.

Inference The process of estimating the probability distribution of latent states or unknown variables from observed data.

Latent State A variable that is not directly observed but is inferred from data.

Likelihood The probability of the observed data under a given model.

Maximum A Posteriori (MAP) The estimate of the most likely latent state, given the observed data. It is the state that maximizes the posterior distribution.

Mixed Traffic A traffic situation where both autonomous vehicles and human-driven vehicles operate together.

Observation The data that is directly measured from the system, represented as feature vectors.

Posterior The probability distribution over the latent states after observing the data.

Precision The ratio of true positive predictions to the total predicted positives. It indicates the accuracy of positive predictions.

Prior The probability distribution over the latent states before observing any data.

Recall The ratio of true positive predictions to the total actual positives. It indicates the completeness of positive predictions.

Random Variable (RV) A variable that can take on different values, each with an associated probability.

State Transition Probability of transitioning from one latent state to another.

Time Headway (THW) The time needed for a vehicle to reach the current position of the vehicle ahead if it maintains its current velocity.

Time to Collision (TTC) The time needed for a vehicle to collide with the vehicle ahead if both of them maintain their current velocity.

Trajectory The time-ordered sequence describing the movement of a vehicle over time.

Kurzfassung

Künstliche Intelligenz (KI) verändert autonome Systeme durch bedeutende Fortschritte in den Bereichen Sensorik, Kommunikation und Rechenleistung. Obwohl vollständig autonome Fahrzeuge (Autonomous Vehicles (AVs)) voraussichtlich immer häufiger werden, werden sie auf absehbare Zeit gemeinsam mit von Menschen gesteuerten Fahrzeugen (Human-Driven Vehicles (HDVs)) am Straßenverkehr teilnehmen. Um in diesem gemischten Verkehrsumfeld sicher zu operieren, müssen autonome Systeme in der Lage sein, menschliche Fahrhandlungen zu modellieren und vorherzusagen.

Diese Arbeit adressiert dieses wichtige vorgelagerte Problem durch die Entwicklung eines transparenten, probabilistischen Rahmens zur Vorhersage hochrangiger menschlicher Fahrhandlungen unter Verwendung dynamischer Bayes'scher Netze (Dynamic Bayesian Networks (DBNs)). Ziel dieser Forschung ist die Entwicklung eines Modells, das zukünftige Fahreraktionen abschätzen kann und gleichzeitig Unsicherheiten in einer interpretierbaren Form darstellt. Ein solches Modell kann innerhalb eines AV-Systems als Belief-Modell dienen und Informationen für nachgelagerte Planungs- und sicherheitsrelevante Module bereitstellen.

Die Studie verwendet den naturalistischen Autobahnfahrdatensatz highD, um zeitliche Muster menschlichen Fahrverhaltens zu erlernen. Fahrzeugtrajektorien werden in sequenzielle, fensterbasierte Beobachtungen umgewandelt, die relevante Bewegungs- und Interaktionsinformationen über die Zeit erfassen. Diese Beobachtungen fassen zusammen, wie sich ein Fahrzeug und der umgebende Verkehr über kurze Zeitintervalle entwickeln. Anschließend wird ein DBN trainiert, um die zeitliche Entwicklung des Fahrerverhaltens zu modellieren und Aktionswahrscheinlichkeiten aus eingehenden Beobachtungen abzuleiten. Das vorgeschlagene DBN verwendet eine hierarchische latente Struktur, in der Aktionen von Fahrregimen abhängig sind. Genauer gesagt repräsentiert das Modell zwei latente Regime, ein Regime mit geringer Interaktion und ein Regime mit hoher Interaktion, und sagt Aktionszustände innerhalb jedes Regimes vorher. Dadurch kann das Modell sowohl das vorhergesagte Fahrmanöver als auch den Interaktionskontext darstellen, in dem es auftritt.

Eine zentrale Stärke dieses Ansatzes liegt in seiner transparenten probabilistischen White-Box-Struktur. Im Gegensatz zu vielen Black-Box-Lernmethoden erzeugt das DBN seine Vorhersagen über explizite probabilistische Beziehungen, die mithilfe der Wahrscheinlichkeitstheorie interpretiert werden können. Dies ist insbesondere für sicherheitsrelevante Anwendungen von Bedeutung, bei denen Interpretierbarkeit und die Darstellung von Unsicherheiten wichtig sind. Das Modell unterstützt daher nicht nur die Vorhersageleistung, sondern auch die Erklärbarkeit, was bei der Betrachtung solcher

Systeme für sicherheitskritische Verkehrsanwendungen von Vorteil ist. Die Studie leistet einen Beitrag zur Forschung im Bereich des autonomen Fahrens, indem sie Aktionsvorhersage, zeitliche probabilistische Schlussfolgerung und Interpretierbarkeit in einem einheitlichen Rahmen kombiniert.

Die Ergebnisse zeigen, dass das Modell in der Lage ist, aussagekräftige Verhaltensmuster aus naturalistischen Verkehrsdaten zu lernen und vielversprechende Manövervorhersagen über einen zukünftigen Zeithorizont zu generieren. In der aktuellen Evaluation erreichte das Modell eine exakte Next-Step-Genauigkeit von 46,4 %, eine Makropräzision von 44,3 % sowie eine Trefferquote von 68,0 % über einen Vorhersagehorizont von zwei Sekunden. Eine Vorhersage zum Zeitpunkt t wird dabei als Treffer gezählt, wenn das vorhergesagte Manöver innerhalb der nächsten fünf Fensterschritte auftritt, was etwa zwei Sekunden entspricht. Die durchschnittliche Time-to-Hit betrug 0,50 s und der Median 0,40 s, was zeigt, dass viele korrekte Vorhersagen früh innerhalb des Vorhersagehorizonts auftreten. Diese Ergebnisse deuten darauf hin, dass das Modell zukünftige menschliche Fahreraktionen innerhalb eines praktisch nutzbaren kurzfristigen Zeitintervalls antizipieren kann, anstatt nur einen einzelnen exakten zukünftigen Zeitpunkt vorherzusagen. Implementierungs-Benchmarks zeigen zudem geringe Online-Inferenzkosten mit einer mittleren CPU-Latenz von 0,164 ms pro Fahrzeug für Belief-Update und nächste Aktionsvorhersage.

Insgesamt zeigen die Ergebnisse, dass DBNs ein vielversprechender Ansatz für die probabilistische Vorhersage menschlicher Fahreraktionen sind. Die Studie demonstriert, dass hochrangige menschliche Fahrmanöver mithilfe naturalistischer Verkehrsdaten in einer interpretierbaren und zeitlich strukturierten Weise modelliert werden können. Dies stellt einen wichtigen Schritt hin zu einem besseren Verständnis menschlichen Fahrverhaltens im gemischten Verkehr dar und kann AV-Systeme dabei unterstützen, das Verhalten umliegender menschlicher Fahrer besser einzuschätzen.

Stichwörter: Vorhersage menschlichen Fahrverhaltens, Autonomes Fahren, Von Menschen gesteuerte Fahrzeuge, Dynamische Bayes’sche Netze, Manövervorhersage, Probabilistische Modellierung, Gemischte Verkehrsumgebungen

Abstract

Artificial Intelligence (AI) is transforming autonomous systems through major advances in sensing, communication, and computation. Although fully Autonomous Vehicles (AVs) are expected to become more common, they will share the road with Human-Driven Vehicles (HDVs) for the foreseeable future. To operate safely in this mixed-traffic setting, autonomous systems must be able to model human driver behavior and predict future maneuvers.

This work addresses this important upstream problem by developing a transparent, probabilistic framework for predicting high-level human driving behavior using Dynamic Bayesian Networks (DBNs). The purpose of this research is to develop a model that can estimate likely future maneuvers while also representing uncertainty in an interpretable form. Such a model can act as a belief model inside an AV framework and provide information for downstream planning and safety-related modules.

The study uses the highD naturalistic highway driving dataset to learn temporal patterns of human driving behavior. Vehicle trajectory data are converted into sequential window-based observations that capture relevant motion and interaction information over time. These observations summarize how the vehicle and its surrounding traffic evolve over short time intervals. A DBN is then trained to model the temporal evolution of driver behavior and to infer action probabilities from incoming observations. The proposed DBN uses a hierarchical latent structure in which actions are conditioned on driving regimes. More specifically, the model represents two latent regimes, low-interaction and high-interaction, and predicts discrete action states within each regime. This allows the framework to represent both the predicted maneuver and the interaction context in which it occurs.

A key strength of this approach is that it provides a transparent, white-box probabilistic structure. Unlike many black-box learning methods, the DBN makes predictions through explicit probabilistic relationships that can be interpreted using probability theory. This is valuable in safety-relevant applications, where interpretability and uncertainty representation are important. The model therefore supports not only prediction, but also explainability, which is useful when such systems are considered for use in safety-critical transport settings. The study contributes to science and engineering by combining maneuver prediction, temporal probabilistic reasoning, and interpretability in a single framework for autonomous driving applications.

The results show that the model can learn meaningful behavioral patterns from naturalistic traffic data and generate promising maneuver predictions over a future horizon. In the current evaluation, the model achieved 46.4% exact next-step accuracy, 44.3% macro precision, and a 68.0% hit rate over a 2-second prediction horizon. Here, a prediction made at time t is counted as a hit if the predicted maneuver appears at any point within the next 5 window steps, corresponding to about 2 seconds. The mean time-to-hit was 0.50 s, and the median time-to-hit was 0.40 s, showing that many correct predictions occur early within the horizon. These results suggest that the model can anticipate future human driver maneuvers within a practically useful short-term interval rather than only at one exact future instant. Implementation benchmarks also indicate low online inference cost, with a mean single-vehicle Central Processing Unit (CPU) latency of 0.164 ms for belief update and next-maneuver prediction.

Overall, the findings indicate that DBNs are a promising framework for probabilistic human driver maneuver prediction. The study shows that high-level human driving maneuvers can be modeled in an interpretable and temporally structured way using naturalistic traffic data. This provides a useful step toward better behavior understanding in mixed traffic and may support AV systems that need to reason about nearby human drivers.

Keywords: Human Driver Behavior Modeling, Autonomous Driving, Human-Driven Vehicles, Dynamic Bayesian Networks, Maneuver Prediction, Probabilistic Modeling, Mixed Traffic Environments

1. Introduction

Automation is influencing almost every part of daily life, and transportation is one of the areas changing most rapidly. Autonomous Vehicles (AVs) are at the forefront of this change, revolutionizing the transportation sector. Their development has accelerated because of recent progress in Artificial Intelligence (AI), sensing systems, computing, and communication technologies. These innovations could enhance traffic efficiency, increase travel comfort, improve road safety, and, most importantly, reshape future mobility.

However, fully autonomous traffic systems do not yet exist anywhere in the world. AVs will have to operate in mixed traffic together with Human-Driven Vehicles (HDVs). This creates important challenges because different road users provide different types and amounts of information. While AVs can rely on rich on-board sensing and may also use Vehicle-to-Everything (V2X) communication [1], the intentions of human drivers are not directly observable and must be inferred from their behavior in traffic [2]. Consequently, their intentions and decisions cannot be measured directly, but instead have to be estimated from observable cues such as vehicle motion, lane changes, relative positions, and interactions with nearby vehicles.

For this reason, human driver behavior prediction is an important problem in autonomous driving. If an AV can estimate what nearby human drivers are likely to do next, it can make safer and better-informed decisions. Anticipating likely braking, lane-changing, or car-following behavior can support downstream tasks such as risk assessment, trajectory prediction, and motion planning [2], [3], [4].

Over the last two decades, researchers have developed three broad families of behavioral models for road users: rule-based models, probabilistic models, and learning-based methods [3], [5], [6]. In parallel, several game-theoretic decision-making paradigms for AVs have emerged [7]. Rule-based models are simple and easy to understand, but they are often too rigid to capture the variability of real human behavior. Learning-based methods can learn complex patterns from large datasets, but many of them are difficult to interpret and are considered as black boxes. Probabilistic models provide a useful middle ground because they can explicitly represent uncertainty, capture temporal dependencies, and still remain interpretable.

This thesis focuses on probabilistic modeling of high-level human driving behavior in highway traffic. Rather than building the full control or planning stack of an AV, this work addresses an important upstream problem: predicting the likely future maneuvers of surrounding human drivers. The goal is to develop a model that can estimate likely future maneuvers while also representing uncertainty in a clear and interpretable way. In this thesis, future maneuvers are represented internally as discrete action states.

To achieve this, the thesis uses a Dynamic Bayesian Network (DBN) to model the temporal relationship between observable driving features and hidden behavioral states. Since driver intentions are not directly observable, the model represents them through latent variables and infers them probabilistically from trajectory-based observations [8]. In particular, the DBN is designed to capture latent factors such as driving context and action dynamics, and to produce probabilistic predictions of future high-level maneuvers. The model is trained and evaluated using the highD dataset, a naturalistic highway driving dataset containing detailed vehicle trajectory recordings [9]. From these trajectories, relevant features are extracted to describe motion, surrounding traffic context, and interaction behavior. These features serve as observations for the DBN.

The thesis studies whether such a probabilistic framework can provide accurate, interpretable, and computationally efficient short-horizon predictions of future human driving maneuvers. In particular, it examines the suitability of the dataset, the structure of the DBN, the usefulness of the selected observation features, the quality of future maneuver prediction, the interpretability of the learned states, and the runtime performance of the model for online use.

2. Theoretical Background

2.1. Probability Theory

Human driving behavior is inherently stochastic. In a traffic scenario, different drivers may interpret the same situation differently depending on their perception, experience, and intentions. As a result, multiple maneuvers may be possible under the same traffic conditions. For example, a driver approaching a slower vehicle may choose to brake, maintain speed, or initiate a lane change. Since such decisions cannot be determined deterministically, behavior prediction must account for uncertainty. Probability theory provides a mathematical framework for representing and reasoning about uncertainty. Instead of predicting a single deterministic outcome, probabilistic models represent a distribution over possible outcomes and their likelihoods. This allows the model to capture the variability and uncertainty inherent in human decision-making.

In probability theory, each uncertain quantity is represented as a Random Variable (RV) [10]. In the context of driver behavior modeling, variables such as vehicle velocity, acceleration, or lane position can be treated as observable RVs. In addition, latent quantities such as driver style or driving maneuver intention can also be represented as RVs that must be inferred from observations.

Complex systems require modeling interactions between many RVs. When several RVs are considered together, their behavior can be described using a joint probability distribution [10]. For two RVs, O and Z, representing the observable RV and latent RV, the joint probability is written as $P(O, Z)$ and represents the probability that both variables take specific values at the same time. This idea can be extended to any number of variables. In driver behavior modeling, joint probability is important because observable variables and latent variables influence each other and must often be described together within a unified probabilistic framework.

It is also important to describe dependencies between RVs. This is done using conditional probability [10]. The conditional probability of O given Z is defined as

$$P(O \mid Z) = \frac{P(O, Z)}{P(Z)},$$

provided that $P(Z) > 0$. Conditional probability describes how the probability of one RV changes when information about another RV is known. This concept is essential because the probability of a future driving maneuver is estimated conditioned on the observed traffic context during inference. During model learning, the probability of observations conditioned on latent states is also required.

Another important concept is marginal probability, which gives the probability of a subset of RVs irrespective of the values of the others [10]. For example, if the joint distribution $P(O, Z)$ is known, the marginal probability of Z can be obtained by summing over all possible values of O:

$$P(Z) = \sum_{O} P(O, Z).$$

In addition to describing dependencies between RVs, it is also useful to identify situations where variables do not influence each other. Two RVs are said to be independent if the value of one variable does not affect the probability of the other. Formally, two variables A and B are independent if their joint probability can be written as

$$P(A, B) = P(A)P(B).$$

In many practical problems, variables are not completely independent but may become independent when conditioned on another variable. This property is known as conditional independence. Two variables A and B are conditionally independent given a third variable C if the joint conditional distribution factorizes as

$$P(A, B \mid C) = P(A \mid C)P(B \mid C).$$

Conditional independence is a key concept in probabilistic graphical models because it allows complex joint probability distributions to be represented using simpler conditional relationships between variables [11].

2.1.1. Bayes' Theorem

A fundamental principle in probabilistic reasoning is Bayes' theorem, which allows beliefs about uncertain events to be updated when new observations become available [10]. It relates conditional and marginal probabilities and is expressed as

$$P(Z \mid O) = \frac{P(O \mid Z)P(Z)}{P(O)} \tag{2.1}$$

Here, $P(Z)$ is called the prior probability. The prior represents the initial belief about a variable before observing any new data. In the context of driver behavior modeling, the prior can represent the initial probability of different latent driver states before considering the current traffic situation. The term $P(O \mid Z)$ is called the likelihood. The likelihood describes how probable the observed data O is, given a particular latent state Z. In driver behavior prediction, it measures how compatible the observed vehicle motion and traffic features are with a given latent driver state. The term $P(O)$ is called the evidence (or marginal likelihood). It represents the total probability of observing the data O under all possible values of Z. The evidence acts as a normalization factor that ensures the resulting probabilities sum to one. Finally, $P(Z \mid O)$ is called the posterior probability. The posterior represents the updated belief about Z after observing O. In behavior modeling, the posterior corresponds to the updated probability distribution over the latent driver state given the observed traffic context.

Bayes' theorem therefore provides a systematic way to update probabilistic beliefs when new information becomes available. This principle forms the basis for probabilistic inference in many models used for behavior prediction, including DBN.

2.2. Probabilistic Graphical Models

Probabilistic graphical models provide an efficient way to represent complex joint probability distributions over many RVs [10], [11]. As the number of variables increases, modeling the full joint distribution becomes difficult because the dependencies between variables also increase. Probabilistic graphical models address this problem by representing RVs and their probabilistic relationships using a graph. In this representation, nodes correspond to RVs, while edges describe dependencies between them. The graph structure also encodes conditional independence assumptions. These assumptions allow the joint probability distribution to be factorized into simpler Conditional Probability Distributions (CPDs).

There are two main types of probabilistic graphical models:

1. **Directed graphical models**, where edges have a direction and represent conditional dependencies between variables.

2. **Undirected graphical models**, where edges do not have a direction and represent symmetric relationships between variables.

One important class of probabilistic graphical models is the Bayesian Network (BN). Its temporal extension is the DBN, which is particularly suitable for modeling sequential processes over time.

2.2.1. Bayesian Networks

A BN is a directed probabilistic graphical model that represents a set of RVs and their conditional dependencies using a Directed Acyclic Graph (DAG) [11]. Each variable in the network is associated with a CPD, which describes how the variable depends on its parent variables. These CPDs are often represented using Conditional Probability Tables (CPTs). If a variable has no parents in the graph, its distribution is given by a prior probability distribution.

In general, the joint probability distribution of all variables in a BN can be factorized according to the graph structure. For a set of variables $X_1, X_2, \dots, X_n$, the joint distribution can be written as

$$P(X_1, X_2, \dots, X_n) = \prod_{i=1}^{n} P(X_i \mid \mathrm{Pa}(X_i)) \tag{2.2}$$

Here, $\mathrm{Pa}(X_i)$ denotes the set of parent variables of X_i in the graph. This factorization reflects the conditional independence assumption embedded in the network structure.

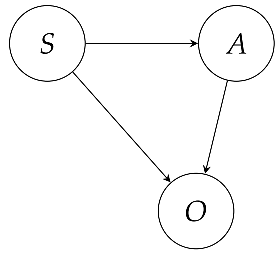


Figure 2.1.: Example of a BN

For the example network shown in Figure 2.1, the joint distribution can be written as

$$P(S, A, O) = P(S)P(A \mid S)P(O \mid S, A).$$

Standard BNs describe relationships between variables within a single time step. However, in many real-world problems, such as driver behavior modeling, the system evolves over time. In such cases, a temporal extension of BNs, called a DBN, is used.

2.2.2. Dynamic Bayesian Networks

In a DBN, the system is represented as a sequence of time-indexed RVs that describe how the system evolves over time. A common representation of a DBN uses a two-slice temporal model that captures dependencies both within a single time step and across consecutive time steps. This representation typically relies on the first-order Markov assumption, which states that the current state depends only on the previous state and not on earlier states.

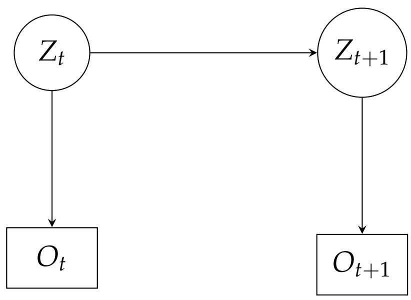


Figure 2.2.: Two-slice representation of a DBN.

In the example shown in Figure 2.2, let Z_t denote the latent state at time step t and O_t denote the corresponding observation. Under the Markov assumption, the joint probability distribution over a sequence of states and observations can be written as

$$P(Z_{1:T}, O_{1:T}) = P(Z_1) \prod_{t=2}^{T} P(Z_t \mid Z_{t-1}) \prod_{t=1}^{T} P(O_t \mid Z_t). \tag{2.3}$$

In this formulation, $P(Z_t \mid Z_{t-1})$ represents the state transition model, which captures how the latent state evolves over time, while $P(O_t \mid Z_t)$ represents the emission model that relates the hidden state to the observed data.

In the context of driver behavior modeling, DBNs provide a probabilistic framework for representing latent driver states and their temporal evolution based on observed vehicle motion and traffic context.

2.3. Inference in Chain-Structured DBNs

Inference in probabilistic graphical models refers to computing the probability distribution of unknown variables given observed evidence. In temporal models such as DBNs, this involves estimating the distribution of latent states from a sequence of observations [10], [11].

For the DBN shown in Figure 2.2, the latent variable Z_t may represent hidden quantities such as driver style or maneuver state. These variables are not directly observable and must therefore be inferred from measurable quantities such as vehicle motion and the surrounding traffic context. Inference is required because the model contains hidden states, whereas only the observation sequence $O_{1:T}$ is available.

In temporal probabilistic models, three common inference tasks arise [12]:

1. **Filtering:** Estimating the current latent state using observations up to the current time step, $P(Z_t \mid O_{1:t})$.
2. **Smoothing:** Estimating the latent state using the full observation sequence, $P(Z_t \mid O_{1:T})$.
3. **Prediction:** Estimating future latent states from the observations available up to the current time, $P(Z_{t+k} \mid O_{1:t})$.

For the chain-structured DBN shown in Figure 2.2, the latent state at time t depends only on the previous state Z_{t-1}, and the observation O_t depends only on the current state Z_t. This structure corresponds to a first-order Markov model with conditionally independent observations, which is equivalent to the structure of a Hidden Markov Model (HMM). Therefore, exact inference can be performed using the forward-backward algorithm [13]. This algorithm is widely used in HMMs, which are a special case of DBNs.

2.3.1. Forward–Backward Algorithm

The Forward–Backward algorithm computes posterior probabilities of latent states by combining information from past and future observations [13]. More specifically, it estimates the posterior distribution of the hidden state at each time step given the complete observation sequence.

Let the observation sequence be $O_{1:T} = \{O_1, O_2, \ldots, O_T\}$ and the corresponding latent states be $Z_{1:T}$. The main objective is to compute the posterior distribution, $P(Z_t \mid O_{1:T})$, which is called the smoothing distribution.

The Forward–Backward algorithm performs this computation using two recursive quantities: the forward messages and the backward messages.

Forward Pass

The forward variable $\alpha_t(z_t)$ represents the joint probability of observing the partial sequence up to time t and being in state z_t:

$$\alpha_t(z_t) = P(O_{1:t}, Z_t = z_t) \tag{2.4}$$

It can be computed recursively as

$$\alpha_t(z_t) = P(O_t \mid Z_t = z_t) \sum_{z_{t-1}} P(Z_t = z_t \mid Z_{t-1} = z_{t-1}) \alpha_{t-1}(z_{t-1}) \tag{2.5}$$

with initialization

$$\alpha_1(z_1) = P(Z_1 = z_1)\, P(O_1 \mid Z_1 = z_1)$$

Here, $P(Z_t = z_t \mid Z_{t-1} = z_{t-1})$ is the transition probability, $P(O_t \mid Z_t = z_t)$ is the observation likelihood, and $P(Z_1 = z_1)$ is the initial state distribution.

Backward Pass

The backward variable $\beta_t(z_t)$ represents the probability of observing the remaining sequence from time $t+1$ to T given the current state:

$$\beta_t(z_t) = P(O_{t+1:T} \mid Z_t = z_t) \tag{2.6}$$

It is computed recursively in the reverse direction as

$$\beta_t(z_t) = \sum_{z_{t+1}} P(Z_{t+1} = z_{t+1} \mid Z_t = z_t) P(O_{t+1} \mid Z_{t+1} = z_{t+1}) \beta_{t+1}(z_{t+1}) \tag{2.7}$$

with initialization

$$\beta_T(z_T) = 1$$

Posterior State Estimation

Once the forward and backward messages have been computed, the posterior probability of the hidden state at time step t can be calculated. This quantity is commonly denoted as $\gamma_t(z_t)$ and represents the probability that the system is in state z_t at time t given the full observation sequence:

$$\gamma_t(z_t) = P(Z_t = z_t \mid O_{1:T})$$

Using the forward and backward variables, this can be written as

$$\gamma_t(z_t) = \frac{\alpha_t(z_t)\beta_t(z_t)}{P(O_{1:T})} \tag{2.8}$$

where $P(O_{1:T})$ is the likelihood of the full observation sequence. Using the forward and backward variables, it can be written as

$$P(O_{1:T}) = \sum_{z_t} \alpha_t(z_t)\beta_t(z_t).$$

for any time step t.

The Forward–Backward algorithm therefore combines information from earlier observations through the forward pass and from later observations through the backward pass.

Posterior Transition Probability

In addition to the posterior probability of individual states, it is often useful to compute the joint posterior probability of two consecutive states. This quantity gives the posterior probability of a transition from state z_t to state z_{t+1} given the full observation sequence:

$$\xi_t(z_t, z_{t+1}) = P(Z_t = z_t, Z_{t+1} = z_{t+1} \mid O_{1:T})$$

Using the forward and backward variables, this can be written as

$$\xi_t(z_t, z_{t+1}) = \frac{\alpha_t(z_t)\, P(Z_{t+1} = z_{t+1} \mid Z_t = z_t)\, P(O_{t+1} \mid Z_{t+1} = z_{t+1})\, \beta_{t+1}(z_{t+1})}{P(O_{1:T})} \tag{2.9}$$

This posterior transition probability is useful in parameter learning, especially in Expectation-Maximization (EM) based training methods [13].

2.4. Parameter Learning in Latent-State Models

A probabilistic model contains several unknown parameters that must be learned from data. For a chain-structured temporal model such as the one shown in Figure 2.2, these parameters can be collectively written as

$$\theta = \{\pi, A, B\},$$

where π denotes the initial state distribution, A is the state transition matrix, and B represents the parameters of the emission model.

If both the observation sequence $O_{1:T}$ and the latent state sequence $Z_{1:T}$ were known, parameter learning could be performed by maximizing the complete-data likelihood

$$\theta^* = \arg\max_{\theta} P(O_{1:T}, Z_{1:T} \mid \theta) \tag{2.10}$$

However, in latent-variable models the state sequence $Z_{1:T}$ is not observed. Therefore, learning must be based only on the observed data, which gives the Maximum Likelihood Estimation (MLE) objective

$$\theta^* = \arg\max_{\theta} P(O_{1:T} \mid \theta) \tag{2.11}$$

Since the latent state sequence is unknown, the likelihood of the observations is obtained by marginalizing over all possible latent state sequences:

$$P(O_{1:T} \mid \theta) = \sum_{Z_{1:T}} P(O_{1:T}, Z_{1:T} \mid \theta) \tag{2.12}$$

As shown in (2.12), the observed-data likelihood is obtained by summing the joint probability over all possible latent state sequences. Because this summation involves many possible latent state sequences, direct maximization of the observed-data likelihood in (2.11) is difficult. To address this problem, the EM algorithm is commonly used [12]. For chain-structured models such as HMMs, this learning procedure is known as the Baum–Welch algorithm [13].

2.4.1. Expectation–Maximization Algorithm

The EM algorithm is an iterative method for parameter estimation in probabilistic models with latent variables. It addresses the difficulty of maximizing the observed-data likelihood by alternating between estimating the latent states and updating the model parameters.

Expectation Step

In the expectation step (E-step), the current parameter estimate is used to compute the posterior distribution of the latent states given the observations. For the chain-structured model considered here, this is done using the Forward–Backward algorithm described by (2.5) and (2.7). The E-step provides the posterior state probabilities

$$\gamma_t(i) = P(Z_t = i \mid O_{1:T})$$

and the posterior transition probabilities

$$\xi_t(i,j) = P(Z_t = i, Z_{t+1} = j \mid O_{1:T}).$$

As shown in (2.8) and (2.9), these posterior quantities represent the probabilities of state occupancy and state transitions conditioned on the full observation sequence.

Maximization Step

In the maximization step (M-step), the model parameters are updated using the posterior quantities computed in the E-step.

For discrete-state chain-structured models, the initial state distribution is updated as

$$\pi_i = \gamma_1(i) \tag{2.13}$$

The transition probabilities can be re-estimated as

$$A_{ij} = \frac{\sum_{t=1}^{T-1} \xi_t(i,j)}{\sum_{t=1}^{T-1} \gamma_t(i)} \tag{2.14}$$

The parameters of the emission model are updated using the observations weighted by the posterior state probabilities. The exact form of this update depends on the chosen emission model.

The E-step and M-step are repeated until convergence. Under standard EM, each iteration does not decrease the observed-data likelihood, and the algorithm converges to a local optimum or stationary point [12].

3. State of the Art

3.1. Introduction

Existing approaches for modeling and predicting human driver behavior can be broadly grouped into rule-based models, probabilistic models, and learning-based models. These families differ in how they represent driving behavior, uncertainty, and interactions with the surrounding traffic environment.

This chapter reviews the main state-of-the-art approaches for human driver behavior modeling and maneuver prediction, with particular attention to their suitability for uncertainty representation and interpretability. Since human driver maneuver prediction is intended to support autonomous driving systems, the chapter also briefly discusses how such predictions are used in downstream decision making, including interaction-aware planning and risk-aware reasoning.

3.2. Rule-Based Models

Rule-based models characterize driving behavior through explicit mathematical or logical rules. These models relate observable variables such as spacing, relative velocity, and desired speed to longitudinal acceleration or lane-change decisions. Their main advantages are interpretability and low computational cost [14]. For this reason, they are widely used in traffic simulation, Advanced Driver Assistance Systems (ADAS), and baseline behavior modeling. However, as they rely on hand-crafted assumptions about driver decision-making, they often represent behavior in a deterministic or simplified manner.

3.2.1. Car-Following Models

Intelligent Driver Model

The Intelligent Driver Model (IDM) is one of the most widely used car-following models [14]. It expresses a vehicle's acceleration as a continuous function of its current velocity, desired velocity, spacing to the leading vehicle, and relative velocity [15]. The formulation also includes parameters related to desired time headway and comfortable braking, which make the model suitable for safety-oriented traffic simulation. Because of its simple structure and physically meaningful parameters, the IDM remains a common reference model in intelligent driving research.

Gipps' Model

Gipps' model is one of the earliest rule-based car-following models. It is based on the idea that a driver chooses a speed that is both desirable and safe with respect to the preceding vehicle [16]. This safety-oriented formulation allows the model to capture conservative driving behavior and ensures collision-free motion under typical traffic conditions.

Optimal Velocity Model and Full Velocity Difference Model

Another classical approach is the Optimal Velocity Model (OVM), in which the driver's acceleration depends on the difference between the current velocity and an optimal velocity determined by the spacing to the leading vehicle [17]. This model captures the driver's intuition to accelerate when the gap ahead increases and decelerate when it decreases. An extension of the OVM is the Full Velocity Difference Model (FVDM), where, instead of only considering the gap between the vehicles, the model also captures the velocity difference between the ego and preceding vehicles when calculating the acceleration [18]. This helps reduce the oscillations observed in the original OVM.

3.2.2. Lane-Change Models

MOBIL Model

The Minimizing Overall Braking Induced by Lane Changes (MOBIL) model is a lane-changing model built upon the IDM to describe lateral driving behavior [14], [19]. It incorporates incentive and safety criteria to evaluate the feasibility of lane changes. The incentive criterion indicates whether a lane change could be advantageous in terms of the driver's acceleration, whereas the safety criterion ensures that the lane change does not cause unnecessary deceleration for surrounding vehicles.

Despite their interpretability and efficiency, rule-based models have limited ability to represent uncertainty and inter-driver variability in complex traffic scenarios. These limitations motivate the use of probabilistic and learning-based approaches.

3.3. Probabilistic Models

Probabilistic models describe driving behavior as a stochastic process, where driving actions are mapped to probability distributions, and represent uncertainty explicitly [20]. In this framework, a driver's decision to accelerate or change lanes is modeled as a probability function that depends on the current traffic context. This approach allows for multiple possible behavioral outcomes, rather than a single deterministic response. This makes them suitable for modeling the variability and ambiguity that are inherent in human driving behavior.

3.3.1. Hidden Markov Models

The HMM is a widely used probabilistic framework for modeling time-varying stochastic processes in which the underlying system states are not directly observable but produce observable outputs [13]. HMM assumes that human driving behavior evolves through a sequence of decision states, such as lane-keeping, braking, and overtaking, that are not directly measurable [21]. These states produce observable vehicle signals like speed, acceleration, and lane position. The objective of the HMM is to infer the most probable state sequence and predict future states from the observed driving data. The main strengths of HMMs are conceptual simplicity and efficient inference. However, HMMs rely on the Markov property, which simplify the model but limit its ability to capture long-term temporal dependencies and complex contextual interactions [22].

3.3.2. Dynamic Bayesian Networks

DBNs extend static BNs to temporal settings and model dependencies among variables across successive time steps [8], [23]. Compared with HMMs, DBNs provide a more flexible representation because the system state can be represented in factored form rather than as a single discrete variable. This allows DBNs to represent multiple interacting hidden and observed variables, which makes them suitable for modeling the joint evolution of driver state, vehicle behavior, and surrounding traffic context [24]. For instance, Młynarczyk et al. (2025) applied BNs to capture causal links between drivers' affective states and behavioral responses [25], while Talluri et al. (2025) employed DBNs to enhance safety standards in automated driving systems through causal reasoning for lane-change prediction and safety assessment [26]. By explicitly modeling dependencies between hidden cognitive states and observable signals, DBNs provide an excellent framework for intention recognition and maneuver prediction in autonomous driving systems [24].

3.3.3. Gaussian-Based Probabilistic Models

Gaussian-based probabilistic models are widely used for modeling continuous-valued driving behaviors such as acceleration or trajectory changes. A single Gaussian distribution is unimodal, whereas driver behavior in a given traffic context may admit multiple plausible future actions or trajectories [27]. To capture such variability, Gaussian Mixture Models (GMMs) represent the joint distribution of relevant variables, such as positions, velocities, headways, or steering angles, as a weighted sum of multiple Gaussian components. For the behavioral modeling of HDVs at the operational level, probabilistic regression techniques such as Gaussian Mixture Regression (GMR) and Gaussian Process Regression (GPR) are often employed to model continuous driving actions, like acceleration or steering angle, for a given traffic context [28], [29]. For example, Zhang et al. use a GMR model to capture feature correlations and temporal dynamics in car-following behavior [28], while Bethge et al. use Gaussian processes to predict human driving behavior and explicitly account for predictive uncertainty [29]. These models provide low-level behavior representations by capturing the stochastic nature of human responses.

3.4. Learning-Based Models

Deep Learning (DL) has enabled powerful data-driven models that predict future trajectories directly from historical motion, map information, and interaction context. Unlike traditional rule-based or probabilistic frameworks that rely on explicit assumptions about driver intent or traffic interactions, these models learn patterns directly from extensive driving data. By making use of large datasets, they can capture complex, nonlinear relationships between driving context and human decisions with high accuracy.

3.4.1. Classical Machine Learning Models

Classical machine learning methods formed the earliest data-driven foundation for maneuver prediction before the advent of modern DL. Algorithms such as k-Nearest Neighbors (kNNs), Support Vector Machines (SVMs), Decision Trees, and Random Forests were widely applied to predict driver intent, classify maneuvers, or recognize driving styles [30]. These approaches rely on manually designed features rather than automatically learned representations, as used in modern DL approaches [31].

3.4.2. Recurrent Neural Network-Based Models

Recurrent Neural Networks (RNNs) form a specialized class of Neural Networks (NNs) designed to model temporal relationships in sequential data [32]. They incorporate internal states and cyclic connections that enable them to process time-dependent information, making them suitable for driver intention recognition and trajectory prediction in behavioral prediction tasks [33], [34]. However, traditional RNNs suffer from the vanishing gradient problem [35]. To address this limitation, the Long Short-Term Memory (LSTM) architecture was introduced. LSTM networks incorporate a dedicated memory cell that helps preserve contextual information over longer time horizons. Although such models perform well in vehicle prediction tasks, they still face limitations in long-horizon forecasting and in handling complex road geometries [33], [36].

Xin et al. (2019) proposed an intention-aware dual LSTM network that separately models the driver's intention and motion dynamics [33]. The first LSTM module captures the latent driving intention using past motion data and environmental context, while the second predicts the future trajectory conditioned on the inferred intention. This dual-stage design allows the model to reason about how inferred driver intentions influence subsequent maneuvers.

Hao et al. (2024) present a hybrid framework that combines a DBN for lane-change intention prediction with a driving-style estimator constructed from K-means clustering and Bayesian filtering [34]. These probabilistic cues are then integrated into an LSTM encoder-–decoder, which produces personalized future trajectories conditioned on both driver intent and style. This fusion allows the network to capture maneuver likelihoods and behavioral variability more effectively than pure LSTM models.

3.4.3. Graph Neural Networks

Modeling the relational structure among traffic participants is important in dynamic traffic scenes. Graph Neural Networks (GNNs) provide an effective framework for representing and learning such multi-agent interactions. In a GNN, each vehicle can be represented as a node, while edges encode the interactions between vehicles [37]. Recent research has explored integrating GNNs with temporal sequence models such as LSTMs and Transformers to jointly capture spatial and temporal dependencies. Abdelraouf et al. (2023) proposed a Temporal Graph Neural Network (T-GNN) that combines Graph Convolutional Networks (GCNs) with LSTM layers to personalize trajectory prediction for different driving styles [38].

However, the performance of GNNs depends strongly on the design of the interaction graph, since graph construction determines which neighboring agents and relations are modeled [37]. An overly sparse graph structure may lead to loss of critical interaction information, while overly dense connections can introduce noise and computational complexity. Moreover, most existing GNN-based trajectory prediction models are highly data-driven and require large, diverse datasets for generalization.

3.4.4. Transformers

Transformers are a recent and powerful DL architecture that are particularly effective at capturing long-range dependencies in sequential data. Unlike recurrent networks, they rely on self-attention mechanisms that directly model relationships between all elements in a sequence, regardless of their temporal distance. In addition, self-attention enables parallel processing of the entire input sequence, which improves computational efficiency [39], [40]. In the multi-head self-attention mechanism, different attention heads can focus on different aspects of the input, and the resulting representations are aggregated to predict future maneuvers or trajectories.

SceneTransformer introduced a model that unified motion prediction and planning through a scene-level attention mechanism that jointly predicts the trajectories of all agents within a traffic scene [41]. Instead of treating each vehicle as an independent entity, SceneTransformer predicts the joint future of multiple interacting agents. Motion Transformer (MTR) extended this approach by introducing a hierarchical encoder–decoder framework that explicitly models behavioral uncertainty, producing multiple possible future trajectories per agent [42]. Building upon these advances, the Hierarchical Vector Transformer (HiVT) introduces a multi-scale hierarchical attention structure designed to handle large numbers of interacting agents efficiently [43]. HiVT first applies local attention to encode trajectory and map features for each agent, then applies a global attention layer to model interactions among all agents in the scene. Using these representations, the decoder predicts the future trajectories of all agents in a single forward pass.

However, despite their strong representation capacity, Transformer-based models generally require large datasets and careful regularization to avoid overfitting. Moreover, their interpretability remains limited compared to probabilistic graphical models or physics-based approaches.

3.4.5. Conditional Variational Autoencoders

Conditional Variational Autoencoders (CVAEs) are probabilistic deep generative models that extend the traditional Variational Autoencoder (VAE) framework by incorporating contextual information into both the encoder and decoder networks [44]. In trajectory prediction, the input observation x captures contextual information while the corresponding output y represents the future trajectory to be predicted. A latent variable z is introduced such that its conditional posterior is approximated through $q_\phi(z \mid x, y)$, which is learned by the encoder.

By conditioning both the encoder and decoder on the contextual variable x, a CVAE can capture complex, multimodal relationships between the observed scene and possible futures, enabling diverse yet context-consistent predictions. The latent variable z captures hidden factors, such as intent or other stochastic influences, that are not directly observable from the input context. As a result, the model can generate a range of possible future trajectories that represent different behavioral modes within the same traffic scenario.

Trajectron++ represents one of the popular CVAE-based frameworks for multimodal trajectory prediction. It combines LSTM-based temporal encoding, graph-structured interaction modeling, and probabilistic latent variable reasoning [45]. Each traffic participant is represented as a node in a dynamically evolving spatiotemporal graph, where edges capture interactions with surrounding agents. The past motion history of each agent is first processed through an LSTM network to encode temporal dependencies, after which a GNN aggregates contextual information across agents. The resulting embeddings are passed into a CVAE decoder, which generates diverse future trajectories by sampling from a learned latent space.

While Trajectron++ focuses on interaction-aware multimodality, Predicted Endpoint Conditioned Network (PECNet) follows a related goal-conditioned approach to trajectory prediction [46]. It models driver intent by predicting a distribution over potential final endpoints, which are spatial points that represent the driver's intended goal. The model then generates complete trajectories conditioned on these sampled endpoints. This two-step design separates high-level intention from detailed motion planning.

More recent research has explored hierarchical extensions of the CVAE framework to reduce long-horizon prediction error and maintain trajectory consistency over time. Single-stage CVAE models can suffer from reduced prediction quality at longer horizons. For example, the Cascaded CVAE addresses this by refining predictions through multiple

CVAE stages, where each stage predicts short-term motion and subsequently extends or corrects it over longer horizons [47].

3.4.6. Diffusion Models

Diffusion models are generative models that learn a data distribution through a forward process that gradually adds noise to training samples and a reverse process that learns to remove this noise [48]. During inference, the model starts from random noise and iteratively denoises it into a future trajectory conditioned on the observed scene context. In trajectory prediction, this allows the model to generate multiple plausible future motions instead of a single deterministic output, which is useful because vehicle behavior is inherently uncertain and often multimodal [49], [50].

TrajDiffuse is a recent development in trajectory prediction that applies diffusion modeling. It generates accurate and diverse single-agent trajectory predictions while maintaining consistency with environmental constraints [51]. MotionDiffuser extends this concept to multi-agent interaction scenarios [52]. It employs a conditional diffusion process to generate permutation-invariant joint trajectories for multiple interacting agents and integrates a Principal Component Analysis (PCA) based dimensionality reduction technique to improve computational efficiency.

3.4.7. Other Learning-Based Behavior Modeling Approaches

Maximum Entropy Inverse Reinforcement Learning (IRL) is a learning-based behavior modeling approach that infers a latent reward function from demonstrations and models behavior as a probabilistic distribution over trajectories [53], [54]. Rather than directly mapping states to actions, IRL aims to recover the latent reward or cost function that a human driver may implicitly optimize during driving. Classical IRL assumes that the expert follows an optimal policy for an unknown reward, without accounting for variations in actual human behavior [55]. However, this can lead to ambiguity, since multiple reward functions may explain the same observed behavior. To address this, Maximum Entropy Inverse Reinforcement Learning (MaxEnt IRL) introduces the principle of maximum entropy [54]. Instead of selecting a single deterministic explanation, it learns the maximum-entropy distribution over trajectories that remains consistent with the demonstrated feature statistics. This makes the model better suited to capturing uncertainty and variability in human driving behavior. Recent work has also applied MaxEnt IRL to autonomous driving trajectory planning and interaction-aware decision making, for example in merging scenarios [56].

3.5. Frameworks for Downstream Autonomous Decision Making

3.5.1. Game-Theoretic Models for Mixed-Traffic Interactions

Level-k Reasoning Level-k models assume that different drivers operate at different reasoning levels. A level-0 driver is nonstrategic and follows a fixed, non-strategic policy like a standard car-following policy. A level-1 driver best responds to the forecasted behavior of level-0 drivers, a level-2 driver responds to level-1 behavior, and so on [57].

Stackelberg Games In Stackelberg games, AV is often modeled as the leader and the other traffic participants as followers [58]. The leader chooses its action while anticipating the followers' best responses. Correspondingly, each follower chooses a strategy that maximizes its own reward given the leader's decision. The resulting solution is commonly formulated in terms of a Stackelberg equilibrium. This framework has been applied, for example, to merging and lane-changing scenarios where a self-driving car interacts with a human driver [59]. For AVs, Stackelberg formulations are well suited because they align with the idea of the AV as a proactive but cautious decision-maker.

Bayesian Games Bayesian games model interactions where agents do not have full knowledge of each other's characteristics, such as driving style, reaction time, or capability level. Each agent maintains probabilistic beliefs over the other agent's "type" and chooses actions that maximize their expected gain under this uncertainty [60]. This framework is appealing to mixed-traffic risk assessment, where behavioral variability and environmental uncertainty play a crucial role.

3.5.2. Partially Observable Markov Decision Process

Although not a game-theoretic model, Partially Observable Markov Decision Processs (POMDPs) are often discussed alongside strategic frameworks because they explicitly reason about hidden intentions and uncertain environment states through belief updates. In a POMDPs, the AV maintains a probability distribution over latent variables, such as driver intent, and updates this belief through Bayesian filtering. Prior work has applied POMDPs to interactive driving scenarios, where accounting for uncertainty in other road users' behavior is essential for safe and anticipatory maneuver planning [61]. However, POMDP-based planners remain computationally demanding, which restricts their use in real-time autonomous driving.

3.5.3. Risk Modeling in Autonomous Decision Making

In autonomous driving, risk is often characterized using criticality indicators that estimate how close a traffic interaction is to becoming unsafe. Many existing approaches rely on surrogate safety measures derived from the relative motion between the ego vehicle and surrounding traffic. In particular, time-based indicators such as Time to Collision (TTC) and Time Headway (THW) remain widely used, especially for collision risk estimation and longitudinal safety assessment [62], [63].

Such indicators are frequently incorporated into planning and collision-avoidance systems as safety constraints or risk terms. However, they are usually most suitable for specific interaction types, such as longitudinal following or merging, and they do not always capture complex multi-vehicle interactions well [62], [63].

To address these limitations, Nyberg et al. (2021) proposed a risk measure derived directly from a formal safety specification rather than relying only on surrogate indicators [64]. Their method computes both the probability that the AV violates a safety rule and the expected severity of that violation under uncertainty. This severity-aware violation measure is then incorporated into the planner, enabling the decision maker to select trajectories that explicitly balance risk and driving progress [64].

3.6. Comparative Assessment of Approaches for HDV Behavior Modeling and Prediction

Rule-based models are simple, interpretable, and computationally efficient. Their parameters have a direct physical meaning, and they are widely used in microscopic traffic simulation and as baselines for evaluating more complex models. However, they typically treat driving as a largely deterministic process and do not account well for the uncertainty and variability inherent in human decision-making. As a result, they are limited in their ability to capture latent intent and behavioral adaptation in complex real-world scenarios.

Probabilistic models address these limitations by treating driver behavior as a stochastic process. They can represent multiple possible maneuvers and quantify uncertainty over future actions. Within this family, DBNs offer a useful compromise between expressive power and interpretability, since they allow richer dependency structures than simpler models such as HMMs while retaining an explicit probabilistic representation of latent variables. This makes them attractive for HDV modeling, especially when uncertainty, partial observability, and belief updating are important. However, probabilistic models still require careful model design and representative data, and simpler variants may rely on restrictive conditional independence or parametric assumptions.

Learning-based models, especially deep neural architectures, currently form a major part of the state of the art in trajectory prediction on large benchmark datasets. They can capture complex nonlinear dependencies, model interactions between multiple agents, and generate multimodal futures through latent-variable or generative formulations. However, these benefits come at a cost. First, they often depend on large, diverse, annotated datasets, which may be limited for the specific mixed-traffic scenarios. Second, their internal representations are often difficult to interpret, making it harder to relate model outputs to driver intent, uncertainty sources, or downstream risk reasoning.

Given the goals of this thesis, probabilistic models, and DBNs in particular, offer the most suitable compromise. They are expressive enough to encode latent driver states and maneuver tendencies, as well as mixed discrete and continuous variables , while remaining interpretable and compatible with Bayesian belief updating and downstream risk-aware reasoning. Rule-based models remain useful as baselines and for scenario construction, whereas DL models provide an important reference for the current trajectory prediction frontier.

4. Methodology

This chapter presents the methodology used to model and predict short-horizon HDV maneuvers in highway traffic. Starting from raw vehicle trajectories in the highD dataset, the proposed pipeline constructs a vehicle-centric representation, derives interaction and risk-related features, aggregates them over short temporal windows, and uses the resulting observation sequences to train and evaluate a hierarchical DBN.

The target of prediction is a discrete maneuver class rather than an exact continuous future trajectory. The model therefore focuses on high-level action categories such as acceleration, braking, lane changes, and stable following. To support this prediction task, the observation space combines ego-motion variables with contextual information from surrounding vehicles and safety-related quantities that describe the current traffic situation.

The remainder of this chapter describes the overall framework, the dataset used in this work, the preprocessing and feature-engineering pipeline, the window-based observation representation, the proposed DBN structure, and the training and inference procedure. It also introduces the prediction-explanation procedure used to provide a local interpretation of the inferred latent state and maneuver prediction.

4.1. Proposed Framework

The proposed framework converts raw highD trajectory data into structured temporal observations for maneuver prediction. Its main stages are data preparation, feature extraction, temporal window aggregation, DBN training, and inference.

The pipeline starts from frame-level trajectories and transforms them into a consistent vehicle-centric representation. Based on the ego vehicle and its local traffic environment, additional features are derived to capture interaction context, lane-related information, and risk-related properties such as THW and TTC. These steps provide a compact description of the current driving situation beyond the ego vehicle alone.

Since maneuver prediction depends on recent temporal context, the frame-level signals are not used directly as isolated observations. Instead, they are aggregated over short sliding windows to form a window-based sequence representation. Each window summarizes recent motion and interaction patterns and serves as one observation for the probabilistic model.

The prediction model is formulated as a hierarchical DBN with two discrete latent variables: a driving-style state and an action state. This structure enables the model to represent both broader traffic-regime differences and maneuver-specific behavioral variation. The engineered window features are linked to the latent states through the emission model, while the temporal evolution of behavior is modeled through the state transitions.

Model parameters are estimated from the training data using an EM-based learning procedure. To avoid information leakage, the windowized sequences are split at vehicle level into training, validation, and test sets before model fitting. After training, the learned model is used to infer the posterior distribution over latent states for new observation sequences and to predict the current maneuver distribution.

An overview of the complete framework is shown in Figure 4.1. The individual stages are described in detail in the following sections.

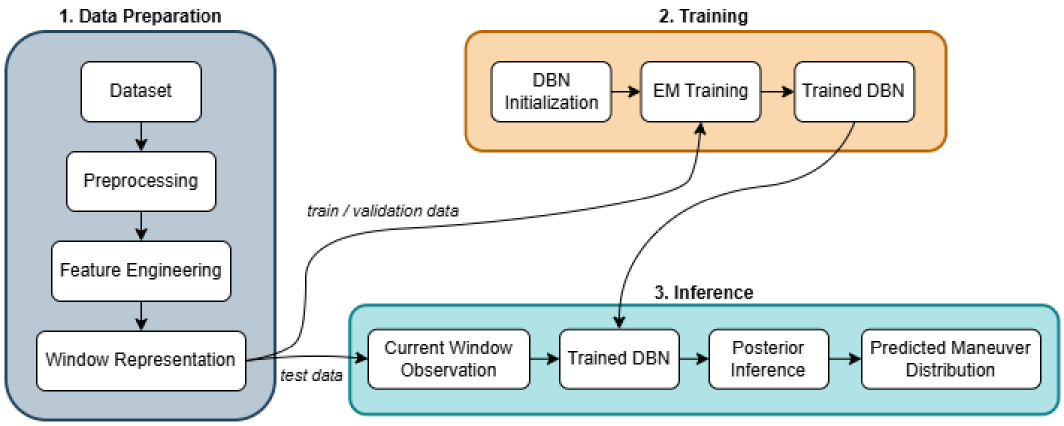


Figure 4.1.: Overview of the proposed framework

4.2. Dataset

This thesis uses the highD dataset [9], a naturalistic highway-traffic dataset recorded from drone videos on German highways. Its aerial perspective provides frame-level trajectories together with surrounding-vehicle information, making it well suited for the analysis of human driving behavior in highway scenarios.

The full highD dataset contains 60 recordings collected at several highway locations. In this work, only the first 25 recordings with three lanes in each driving direction are used. This restriction provides a more uniform road geometry and traffic setting across the selected subset, which supports more consistent behavior modeling. Figure 4.2 shows an example traffic scene from the selected subset and illustrates the multi-lane highway layout together with the surrounding-vehicle context available for each target vehicle.

For each vehicle, highD provides frame-level trajectory variables such as longitudinal and lateral position, velocity, and acceleration. It also includes lane-related information,

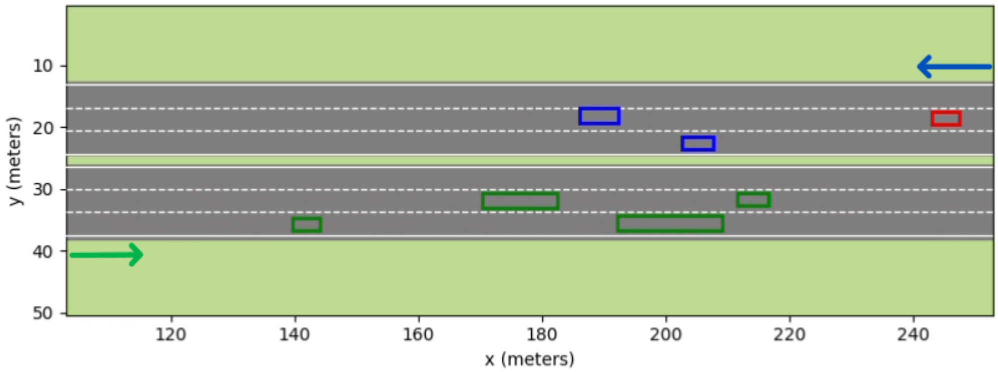


Figure 4.2.: Example traffic scene from a selected highD subset. The red box denotes the ego vehicle, while the blue and green boxes represent other vehicles traveling in the two opposite driving directions.

driving-direction labels, and identifiers for neighboring vehicles in relevant surrounding positions. In addition, the dataset provides traffic-safety indicators such as TTC and THW, which support the construction of interaction and risk-related features. These signals form the basis for the preprocessing and feature-engineering pipeline described in the next sections.

For the selected subset, 50,543 vehicle trajectories are available after loading the chosen recordings. With a window length of $W = 50$ frames, 50,455 trajectories satisfy $T \geq W$, while only 88 are shorter than the required window length. The trajectory lengths are well suited for temporal modeling, with a 10th percentile of 280 frames, a median of 349 frames, and a 90th percentile of 466 frames. This indicates that the selected subset is sufficiently large and temporally rich for constructing window-based observation sequences.

4.3. Data Preparation and Preprocessing

4.3.1. Preprocessing Pipeline

The raw highD files are first merged into a unified per-frame representation that combines trajectory data with vehicle-level and recording-level metadata. Since the dataset contains multiple recordings, each trajectory is uniquely identified by the pair (recording_id, vehicle_id). In addition, vehicle-center coordinates are computed from the original bounding-box positions in order to provide a consistent geometric reference for the subsequent construction of context features.

As the selected recordings contain traffic moving in two opposite roadway directions, the raw image-based coordinates cannot be used directly as a homogeneous representation. A vehicle-centric normalization step is therefore applied using the driving-direction label. This transformation adjusts the sign convention of the longitudinal and lateral motion variables such that forward and lateral motions are represented consistently across both traffic directions. As a result, the learned model receives a unified representation in which the same maneuver follows the same sign convention regardless of the original roadway direction.

At preprocessing stage, the available neighboring positions include the front, rear, left-front, left-side, left-rear, right-front, right-side, and right-rear slots. For each available neighbor, binary existence flags are created and relative kinematic quantities are derived in the form of relative longitudinal distance, relative lateral distance, relative longitudinal velocity, and relative lateral velocity. Missing neighbors remain explicitly marked through the corresponding existence indicators, while the relative features remain undefined when no matching neighbor state is available.

In the final observation vector, only the subset shown in Figure 4.3 is retained. This design focuses the model on the vehicles that most directly describe immediate car-following constraints in the current lane and the local traffic state of the adjacent lanes, while keeping the feature space compact and interpretable. The omitted rear and rear-diagonal neighbors may also contain useful information, but they are excluded here as a simplifying design choice to limit feature dimensionality in the proposed DBN.

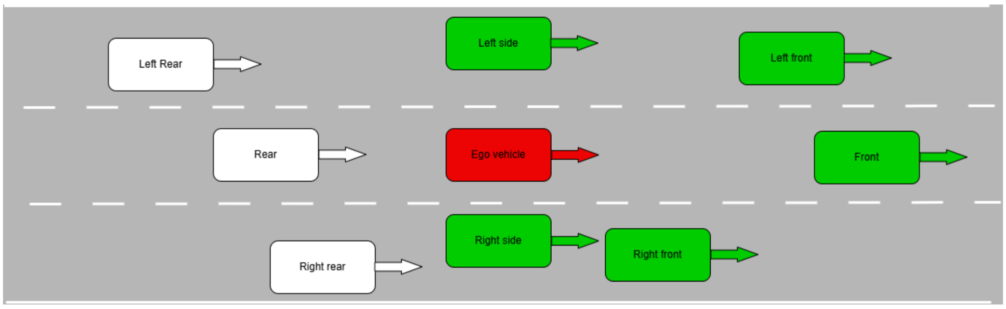


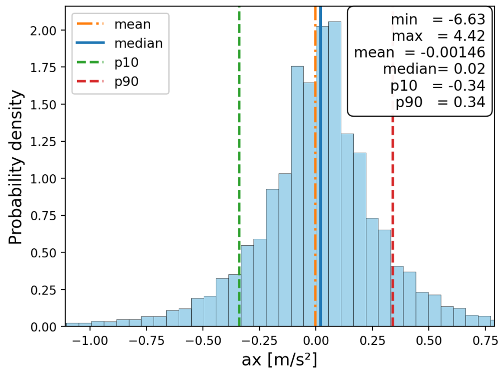
Figure 4.3.: Selected neighbors considered in the observation.

In addition to interaction features, the preprocessing step derives lane-related and risk-related quantities. Distances to the left and right lane boundaries are computed from the recording metadata and the ego-vehicle center position. The lane-change indicator is obtained from changes in lane ID and represented as a ternary variable corresponding to lane change to left, no lane change, and lane change to right. The front-interaction safety signals provided by highD are mapped to the standardized variables `front_thw` and `front_ttc`. Furthermore, ego-vehicle jerk is computed from the frame-to-frame change in longitudinal and lateral acceleration.

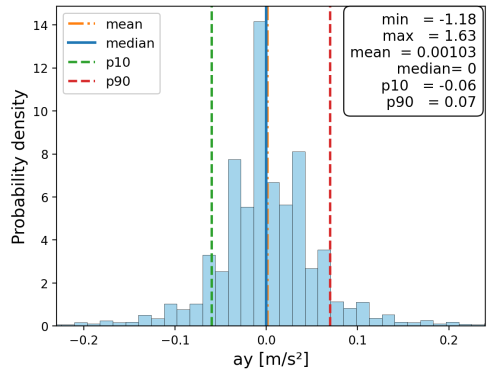
Finally, the processed dataframe is reduced to the selected frame-level observation columns and metadata fields used in the subsequent sequence-construction stage. This produces a consistent per-frame representation that serves as the basis for feature aggregation and windowization.

4.3.2. Data Analysis for Feature Design

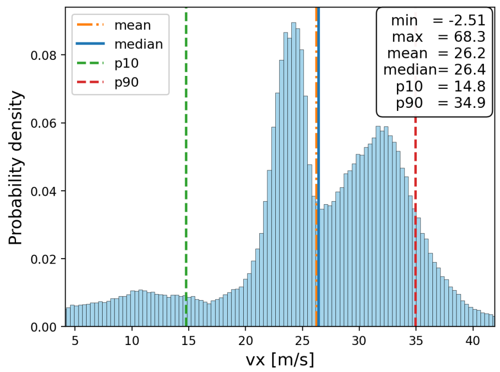
Before defining the final observation vector, an exploratory analysis of the highD varables was carried out to understand the statistical behaviour of the available signals and to identify features that are informative. The analysis focused mainly on kinematic variables, namely longitudinal and lateral acceleration (a_x, a_y), longitudinal and lateral velocity (v_x, v_y), and simple lane-change indicators.



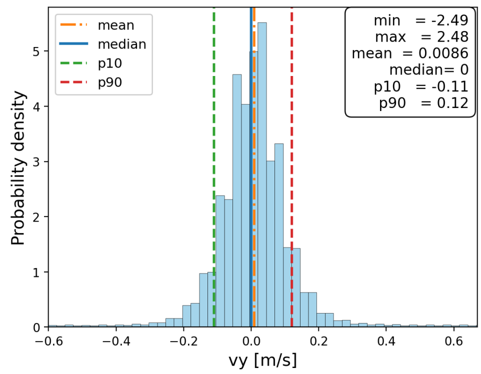
(a) Longitudinal acceleration.



(b) Lateral acceleration.



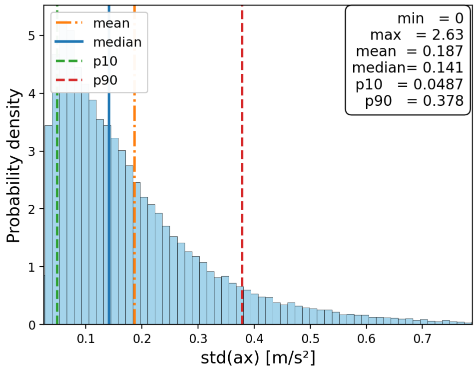
(c) Longitudinal velocity.



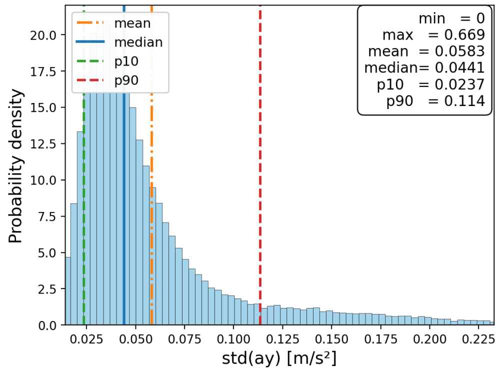
(d) Lateral velocity.

Figure 4.4.: Global distributions of the main kinematic variables.

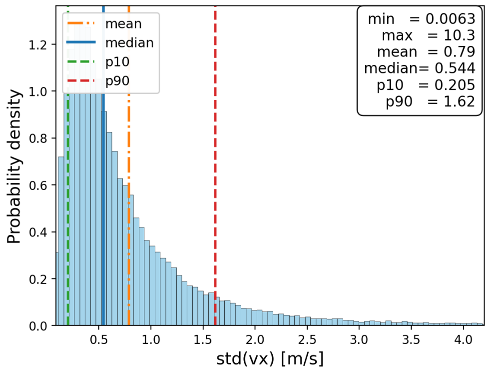
As shown in Figure 4.4a, the distribution of a_x is centred near zero but has visible tails on both sides, indicating that most frames correspond to near-steady motion, while stronger acceleration and braking occur less often. The longitudinal velocity distribution in Figure 4.4c spans a broad range of values and exhibits a multimodal structure. This is mainly caused by differences between vehicle classes, with trucks concentrated at lower speeds and cars occupying a wider and generally higher-speed range. These class-dependent differences show that raw continuous features can have different natural ranges depending on the vehicle type. Class-wise kinematic distributions for cars and



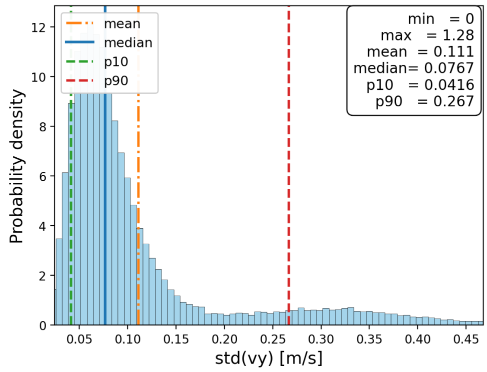
(a) Longitudinal acceleration.



(b) Lateral acceleration.



(c) Longitudinal velocity.



(d) Lateral velocity.

Figure 4.5.: Global trajectory-level variability distributions of the main kinematic variables.

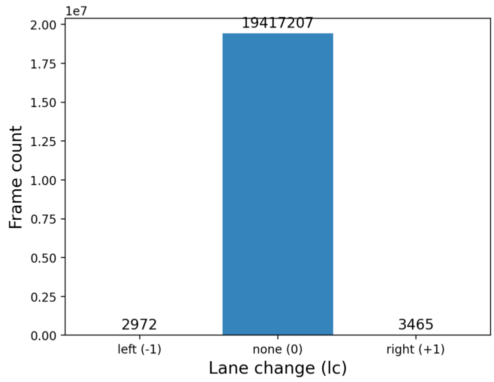
trucks are provided in Appendix A. For this reason, continuous variables are standardized before training, as described later in the training subsection.

On the other hand, the lateral variables are much more concentrated around zero. In Figure 4.4b and Figure 4.4d, both a_y and v_y show sharp peaks near zero, which is expected because most vehicles remain within their lane for most of the time. However, the non-zero tails still reflect lateral adjustments and lane-change-related motion. Together, these observations support the inclusion of both longitudinal and lateral kinematic variables, since the former describe speed adaptation and car-following behaviour, whereas the latter provide cues for lateral maneuvers.

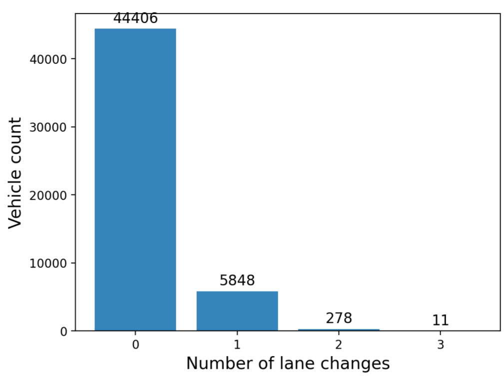
In addition to the raw frame-level signals, the variability of each kinematic variable was also analysed at trajectory level. For each vehicle trajectory, the standard deviation of a_x, a_y, v_x, and v_y was computed over time, producing one variability value per trajectory. Unlike the raw kinematic distributions, these plots do not describe the absolute values of the signals, but rather how strongly they fluctuate within individual trajectories. The resulting distributions are shown in Figure 4.5a to Figure 4.5d.

The variability analysis shows that longitudinal motion is generally more dynamic than lateral motion. In Figure 4.5a and Figure 4.5c, both $\mathrm{std}(a_x)$ and $\mathrm{std}(v_x)$ exhibit broader distributions with long right tails, indicating that some vehicles undergo substantial longitudinal speed and acceleration changes. In contrast, Figure 4.5b shows that $\mathrm{std}(a_y)$ remains concentrated at low values, which is consistent with lateral acceleration being close to zero except during lane changes. A similar pattern appears in Figure 4.5d, although the visible tail suggests that some trajectories contain stronger lateral motion. Class-wise trajectory-level variability distributions are provided in Appendix A. These results support the use of variability-related window features, since instantaneous values alone do not capture whether a trajectory is stable or dynamically changing over time.

The lane-change analysis further highlights an important property of the dataset. As shown in Figure 4.6a, almost all frames belong to the lane-keeping class, while left and right lane changes occur only rarely. The same sparsity appears at trajectory level in Figure 4.6b, where most vehicles perform no lane change and only a small fraction perform one or more. This imbalance shows that lane-change events are very rare compared to straight driving. It also shows that the raw lane-change indicator alone is not sufficient as a learning signal, especially because in highD the lane change is marked only at the frame where the vehicle crosses the lane boundary. This motivated the use of temporal aggregation together with contextual kinematic and interaction features.



(a) Raw lane-change indicator at frame level.



(b) Number of lane changes observed per trajectory.

Figure 4.6.: Lane-change sparsity analysis in the highD data.

Overall, the data analysis led to three main conclusions. First, longitudinal and lateral kinematics provide complementary information for maneuver prediction. Second, trajectory-level variability contains useful information beyond instantaneous frame values. Third, feature scale and class-dependent distribution differences must be handled carefully during preprocessing. These findings directly guided the feature engineering, where the raw signals were transformed into the final observation features for the proposed DBN model.

4.3.3. Feature Engineering

Based on the preprocessing steps described above, the final frame-level observation was defined as the compact feature set listed in Table 4.1. The selected variables describe ego motion, lane-change evidence, lane geometry, local interaction context, and front-interaction risk. This frame-level representation serves as the basis for the subsequent window-based observation model.

Table 4.1.: Frame-level feature groups used in the preprocessing pipeline

Feature group	**Variables**	**Purpose**
Ego kinematics	`vx`, `vy`, `ax`, `ay`, `jerk_x`, `jerk_y`	Describe recent ego motion and control variation
Lane-change evidence	`lc`	Encode left / none / right lane-change evidence
Lane geometry	`d_left_lane`, `d_right_lane`	Represent lateral position within the current lane
Front risk	`front_thw`, `front_ttc`	Describe front-interaction criticality
Neighbor existence	Existence indicators for the selected front and adjacent-lane neighbor slots	Indicate whether relevant surrounding vehicles are present
Relative interaction	For each selected neighbor slot: `dx`, `dy`, `dvx`, `dvy`	Capture relative position and motion with respect to surrounding traffic

4.3.4. Window-Based Representation

The proposed DBN operates on short temporal windows rather than on individual frames. Each vehicle trajectory is therefore converted into a sequence of overlapping windows, as illustrated in Figure 4.7. This allows the model to use recent motion history and interaction context instead of relying only on instantaneous frame values.

Let W denote the window length and S the stride between consecutive windows. In this work, $W = 50$ frames and $S = 10$ frames are used. Since highD is recorded at 25 Hz, each window covers approximately 2 s of history, while consecutive windows start 0.4 s apart. Thus, for a trajectory of length T frames, overlapping windows are extracted every S frames as long as $T \geq W$. Each resulting observation summarizes the selected features over one interval of W consecutive frames.

Each window is converted into a single observation vector by aggregating the frame-level signals over the corresponding window interval. Depending on the variable, the

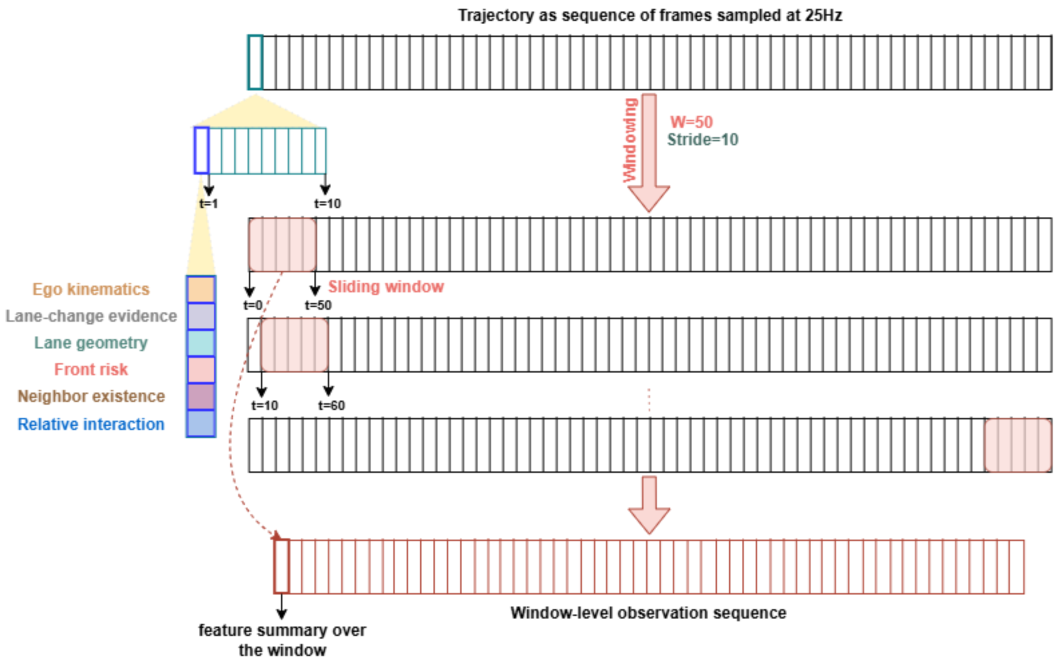


Figure 4.7.: Illustration of the sliding-window representation used in this work.

aggregation uses the most recent value, a minimum over the window, a temporal slope, sign-based fractions, percentile-based summaries, binary event indicators, or fractions of valid or available frames. For neighbor-dependent variables, the summaries are computed only when the corresponding neighbor is present. The complete list of window-level features used in the final observation vector is provided in Appendix B. The resulting sequence of window-level observations forms the input to the proposed DBN during both training and inference.

4.4. Proposed DBN Model

The proposed model represents driver behavior through two discrete latent variables: a style state S_t and an action state A_t. The style state captures a broader driving regime, while the action state represents the maneuver at the current time step. The observation vector O_t corresponds to the window-level features described in the previous section.

4.4.1. Latent State Structure

Rather than using a single latent variable, the model uses a factorized latent representation. As shown in Figure 4.8, the style state evolves over time as its own Markov chain, whereas the action state depends on both the previous action and the current style state.

This structure allows the model to separate slower regime-level dynamics from faster maneuver-level changes.

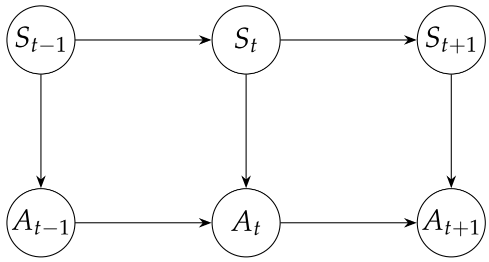


Figure 4.8.: Factorized latent structure of the proposed DBN.

4.4.2. Transition Model

Let S_t denote the latent style state and A_t the latent action state at time step t. The initial distribution is modeled as

$$p(S_0) = \pi_{s_0}, \tag{4.1}$$

$$p(A_0 \mid S_0) = \pi_{a_0|s_0}. \tag{4.2}$$

For later time steps, the transition model is factorized as

$$p(S_t \mid S_{t-1}) = A_s[S_{t-1}, S_t], \tag{4.3}$$

$$p(A_t \mid A_{t-1}, S_t) = A_a[S_t, A_{t-1}, A_t]. \tag{4.4}$$

This factorization reflects the assumption that the style state evolves more gradually, while the current maneuver depends on both the recent maneuver history and the current driving regime.

4.4.3. Emission Model Variants

At each time step, the latent pair (S_t, A_t) generates the window-level observation vector O_t. Two alternative emission formulations are considered in this work, as illustrated in Figure 4.9. Both use the same latent transition structure and differ only in how the observation likelihood is parameterized. In both cases, the observation vector is split into continuous and binary components. The continuous part is modeled with diagonal Gaussian distributions, while the binary part is modeled with Bernoulli distributions.

Product-of-Experts Emission

In the Product of Experts (PoE) formulation, the observation likelihood is decomposed into two experts: one conditioned only on the style state and one conditioned only on the action state. Their contributions are combined multiplicatively and then normalized:

$$p(o_t \mid s_t, a_t) = \frac{p_s(o_t \mid s_t)\, p_a(o_t \mid a_t)}{Z_{s_t,a_t}}, \tag{4.5}$$

where Z_{s_t,a_t} is the normalization term. This formulation allows the style state and the action state to contribute separately to the same observation vector.

Hierarchical Emission

In the hierarchical formulation, a single emission distribution is learned directly for each joint latent-state pair:

$$p(o_t \mid s_t, a_t) = \text{Emission}(o_t; \theta_{s_t,a_t}). \tag{4.6}$$

Here, the observation is modeled jointly for each combination of style and action. This gives the model more direct flexibility at the level of the joint latent state, at the cost of a larger number of emission parameters.

For both emission variants, the final output of the observation model is the log-likelihood tensor

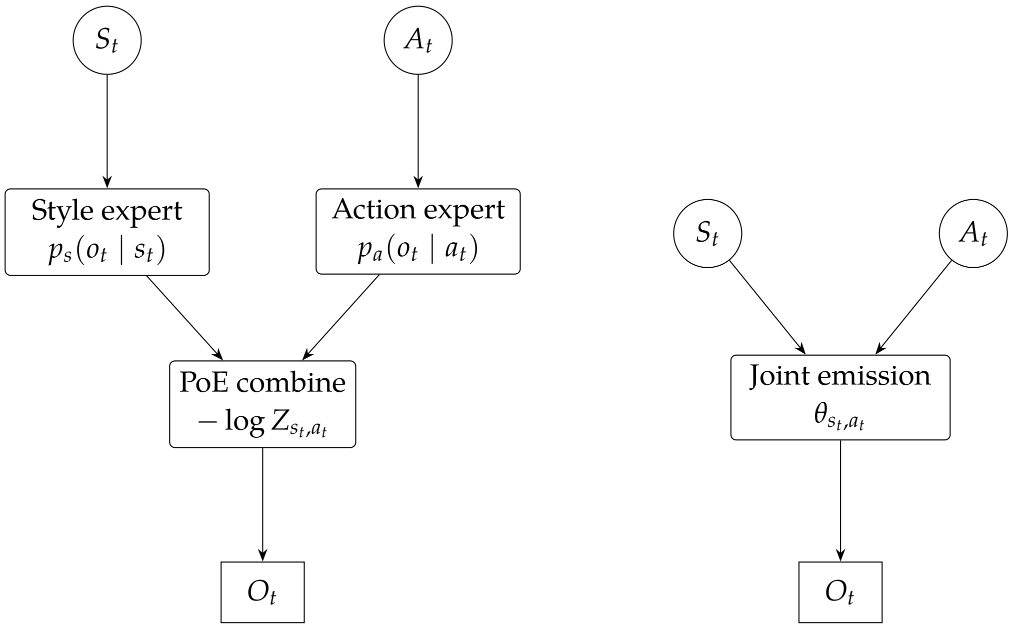
$$\log p(O_t \mid S_t, A_t), \tag{4.7}$$

evaluated for all combinations of style and action states at each time step. These values are then used together with the structured transition model during forward–backward inference and EM training. Thus, the proposed framework keeps the latent dynamics fixed at the DBN level, while allowing different observation models to be compared within the same inference and training pipeline.

4.5. Training and Inference

4.5.1. Prediction Explanation

4.6. Semantic Interpretation of Latent States



(a) Product-of-experts emission.　　(b) Hierarchical emission.

Figure 4.9.: Emission model variants considered in the proposed DBN.

5. Conclusions

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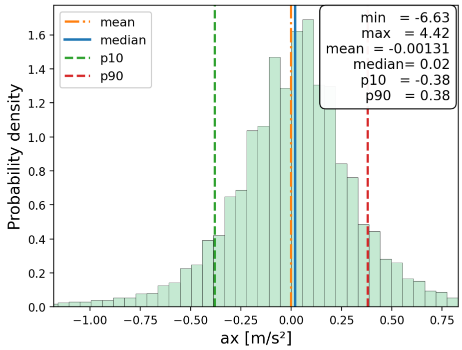
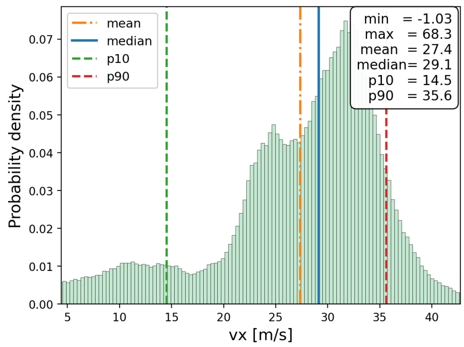
A. Class-Wise Distribution Plots

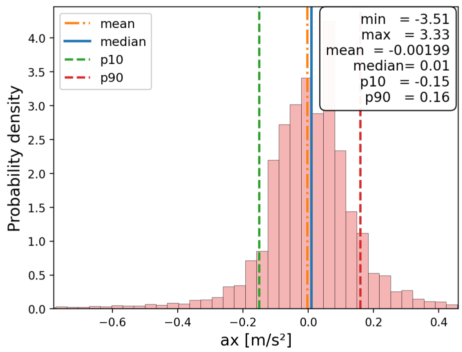
This appendix provides class-wise versions of the main kinematic and trajectory-level variability distributions used in the exploratory data analysis. The plots separate cars and trucks in order to make the class-dependent differences in scale and spread more explicit.



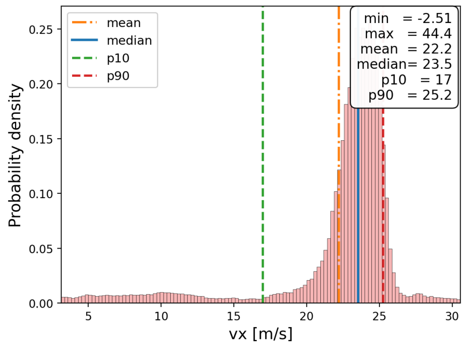
(a) Car: longitudinal acceleration.



(b) Car: longitudinal velocity.

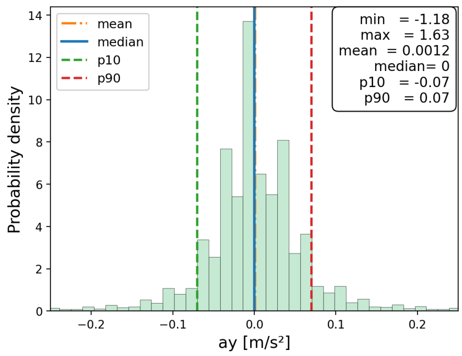


(c) Truck: longitudinal acceleration.

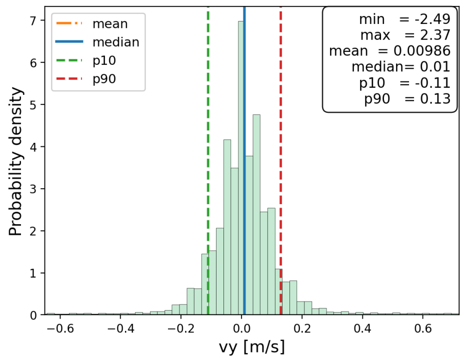


(d) Truck: longitudinal velocity.

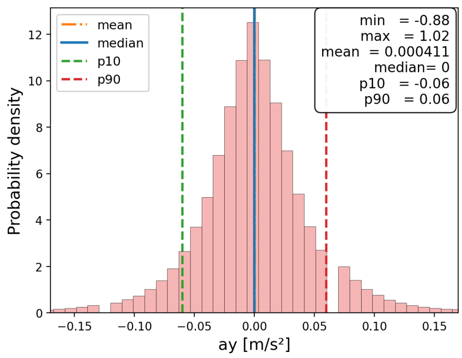
Figure A.1.: Class-wise frame-level distributions of the longitudinal kinematic variables.



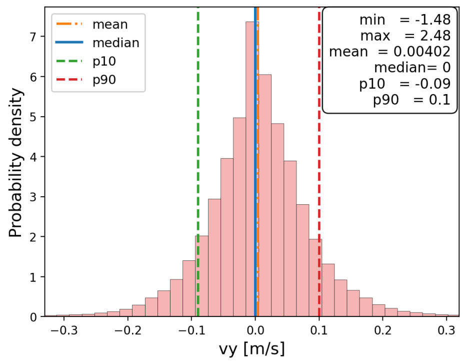
(a) Car: lateral acceleration.



(b) Car: lateral velocity.

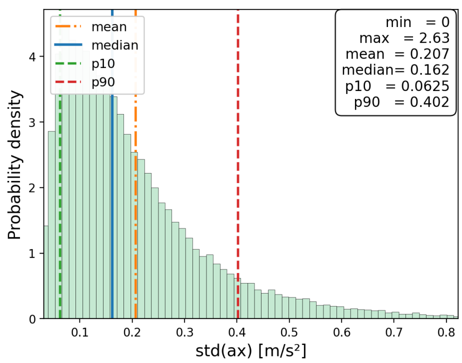


(c) Truck: lateral acceleration.

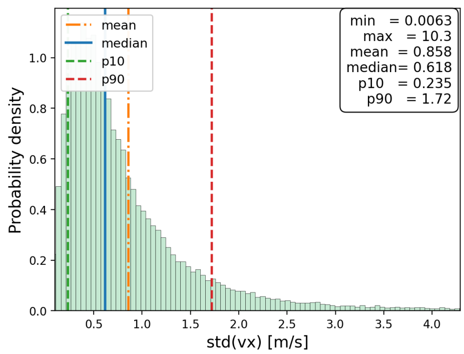


(d) Truck: lateral velocity.

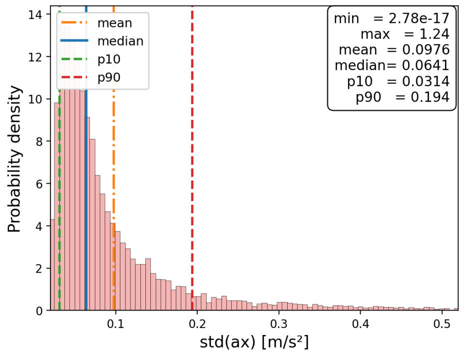
Figure A.2.: Class-wise frame-level distributions of the lateral kinematic variables.



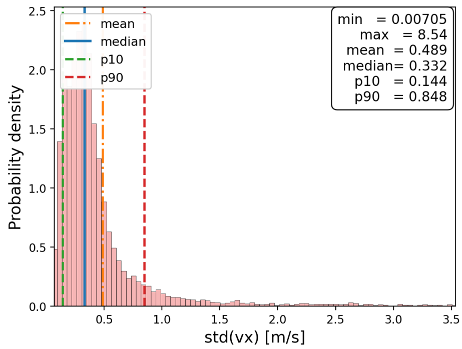
(a) Car: longitudinal acceleration.



(b) Car: longitudinal velocity.

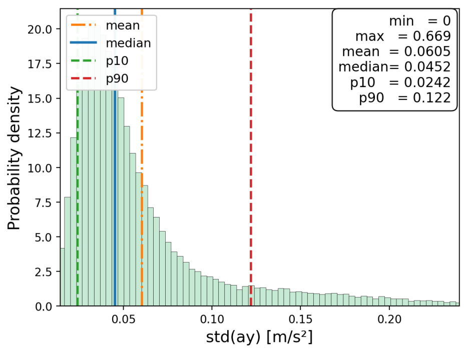


(c) Truck: longitudinal acceleration.

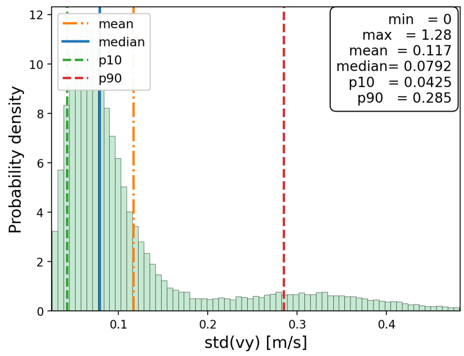


(d) Truck: longitudinal velocity.

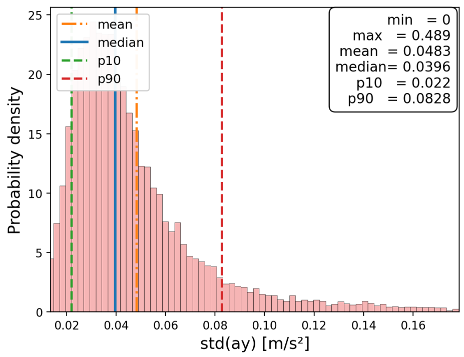
Figure A.3.: Class-wise trajectory-level variability distributions of the longitudinal kinematic variables.



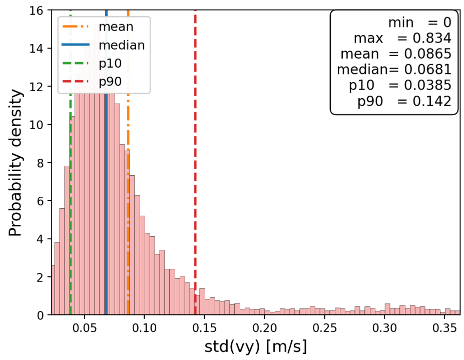
(a) Car: lateral acceleration.



(b) Car: lateral velocity.



(c) Truck: lateral acceleration.



(d) Truck: lateral velocity.

Figure A.4.: Class-wise trajectory-level variability distributions of the lateral kinematic variables.

B. Complete Window-Level Observation Features

This appendix lists the complete set of window-level features used to construct the observation vector of the proposed model. The features are grouped according to their role in the representation.

Table B.1.: Window-level ego-motion features.

Window feature	**Aggregation rule**
`vx_last`	Most recent longitudinal velocity value in the window
`vx_slope`	Temporal slope of longitudinal velocity over the window
`ax_last`	Most recent longitudinal acceleration value in the window
`vy_last`	Most recent lateral velocity value in the window
`vy_slope`	Temporal slope of lateral velocity over the window
`ay_last`	Most recent lateral acceleration value in the window
`ax_neg_frac`	Fraction of frames in the window with negative longitudinal acceleration
`ax_pos_frac`	Fraction of frames in the window with positive longitudinal acceleration
`ax_zero_frac`	Fraction of frames in the window with zero longitudinal acceleration
`ay_neg_frac`	Fraction of frames in the window with negative lateral acceleration
`ay_pos_frac`	Fraction of frames in the window with positive lateral acceleration
`ay_zero_frac`	Fraction of frames in the window with zero lateral acceleration
`jerk_x_p95`	95th percentile of longitudinal jerk within the window
`jerk_y_p95`	95th percentile of lateral jerk within the window

Table B.2.: Window-level lane-change and lane-geometry features.

Window feature	**Aggregation rule**
`lc_left_present`	Binary indicator showing whether a left lane-change event occurs anywhere inside the window
`lc_right_present`	Binary indicator showing whether a right lane-change event occurs anywhere inside the window
`d_left_lane_last`	Most recent distance to the left lane boundary in the window
`d_left_lane_min`	Minimum distance to the left lane boundary within the window
`d_right_lane_last`	Most recent distance to the right lane boundary in the window
`d_right_lane_min`	Minimum distance to the right lane boundary within the window

Table B.3.: Window-level front-risk features.

Window feature	Aggregation rule
`front_thw_last`	Most recent valid THW value within the window
`front_thw_vfrac`	Fraction of frames in the window for which THW is valid
`front_ttc_min`	Minimum valid TTC value within the window
`front_ttc_vfrac`	Fraction of frames in the window for which TTC is valid

Table B.4.: Window-level features related to the front neighbor.

Window feature	Aggregation rule
`front_dx_min`	Minimum relative longitudinal distance to the front vehicle within the window
`front_dvx_slope`	Temporal slope of relative longitudinal velocity with respect to the front vehicle
`front_exists_frac`	Fraction of frames in the window in which a front vehicle is present

Table B.5.: Window-level features related to the side neighbors.

Window feature	Aggregation rule
`left_side_dvx_last`	Most recent relative longitudinal velocity with respect to the left-side neighbor
`left_side_dvx_slope`	Temporal slope of relative longitudinal velocity with respect to the left-side neighbor
`left_side_dy_last`	Most recent relative lateral position with respect to the left-side neighbor
`left_side_exists_frac`	Fraction of frames in the window in which a left-side neighbor is present
`right_side_dvx_last`	Most recent relative longitudinal velocity with respect to the right-side neighbor
`right_side_dvx_slope`	Temporal slope of relative longitudinal velocity with respect to the right-side neighbor
`right_side_dy_last`	Most recent relative lateral position with respect to the right-side neighbor
`right_side_exists_frac`	Fraction of frames in the window in which a right-side neighbor is present

Table B.6.: Window-level features related to the adjacent front neighbors.

Window feature	Aggregation rule
`left_front_dx_last`	Most recent relative longitudinal distance to the left-front neighbor
`left_front_dvx_slope`	Temporal slope of relative longitudinal velocity with respect to the left-front neighbor
`left_front_dy_last`	Most recent relative lateral position with respect to the left-front neighbor
`left_front_exists_frac`	Fraction of frames in the window in which a left-front neighbor is present
`right_front_dx_last`	Most recent relative longitudinal distance to the right-front neighbor
`right_front_dvx_slope`	Temporal slope of relative longitudinal velocity with respect to the right-front neighbor
`right_front_dy_last`	Most recent relative lateral position with respect to the right-front neighbor
`right_front_exists_frac`	Fraction of frames in the window in which a right-front neighbor is present

Declaration of Compliance

I hereby declare to have written this work independently and to have respected in its preparation the relevant provisions, in particular those corresponding to the copyright protection of external materials. Whenever external materials (such as images, drawings, text passages) are used in this work, I declare that these materials are referenced accordingly (e. g. quote, source) and, whenever necessary, consent from the author to use such materials in my work has been obtained.

Signature:

Stuttgart, on the 15 April 2026